西方经济学

Part 4 Business Cycle Theory

The Economy in the Short Run

Lecture 4C The Supply Side

Jian LI

Department of International Economics and Trade
Nanjing University

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Supplement Readings

(1) M12, M13.3; S5, 6.3. [1]

(2) 其他资料：[GM]

- Introduction to Economic Fluctuations
- Aggregate Supply and the Short-Run Tradeoff Between Inflation and Unemployment

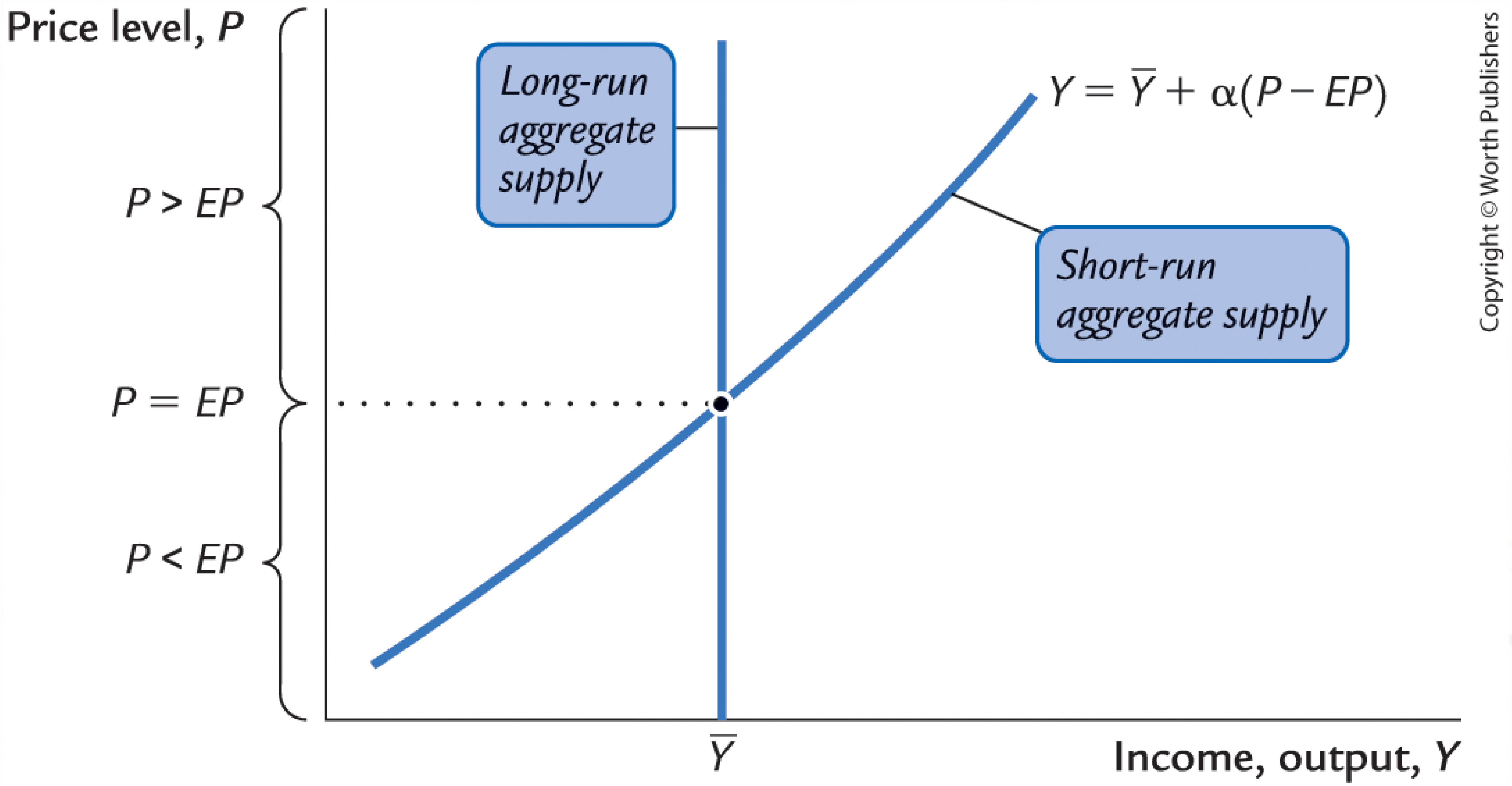
[1] M 指代马工程教材，S 指代课外阅读材料沈坤荣教程。

学习目标

(1) 掌握 AS 曲线的理论。
(2) 掌握凯恩斯主义分析框架：AS-AD。
(3) 理解理性预期在宏观经济政策中的作用。
(4) 领会马工程教材精神。

Outline

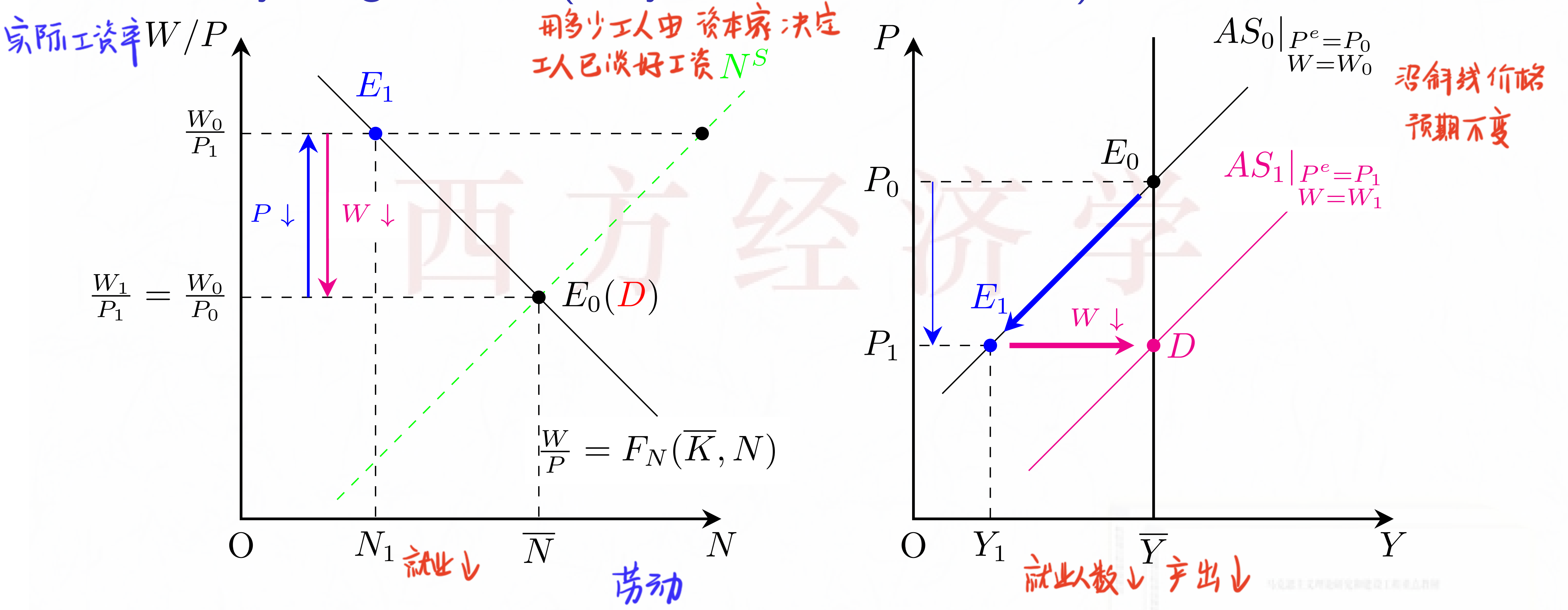
The Shape of Aggregate Supply Curve



$$Y_t = \overline{Y} + \alpha \cdot (P_t - \mathbb{E}_{t-1}[P_t]), \quad \alpha \in [0, +\infty]$$

$$\Rightarrow P_t = \mathbb{E}_{t-1}[P_t] + \frac{1}{\alpha}(Y_t - \overline{Y})$$

The Sticky-Wage Model (Gray, 1976; Fischer, 1977)



名义工资

Setting the nominal wage: $W = w^* \cdot P^e,$

实际工资

The real wage: $\dfrac{W}{P} = w^* \cdot \dfrac{P^e}{P}.$

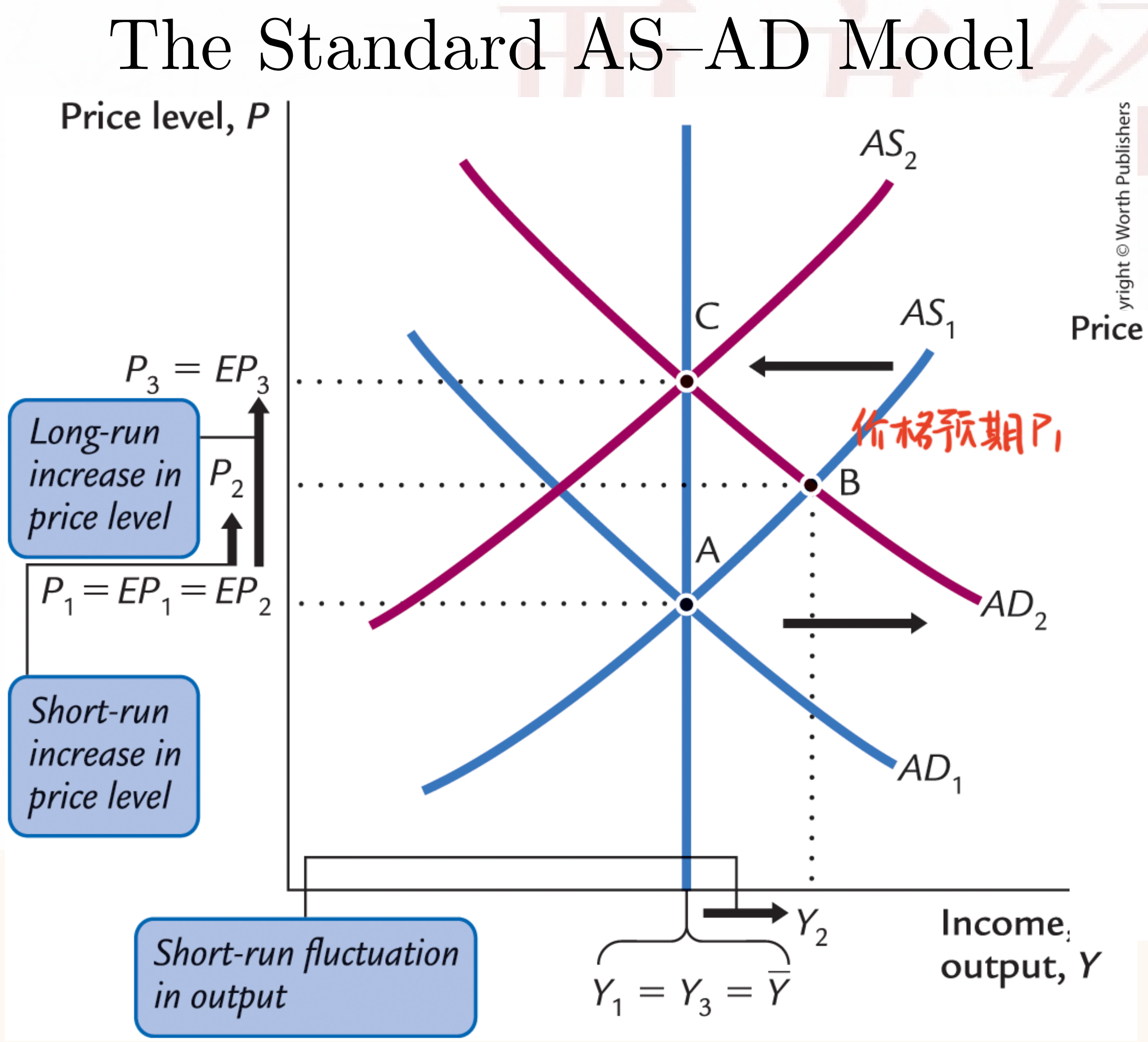
In the short run: $P_0 \searrow P_1 \Rightarrow \overline{N} \searrow N_1, \ \overline{Y} \searrow Y_1.$

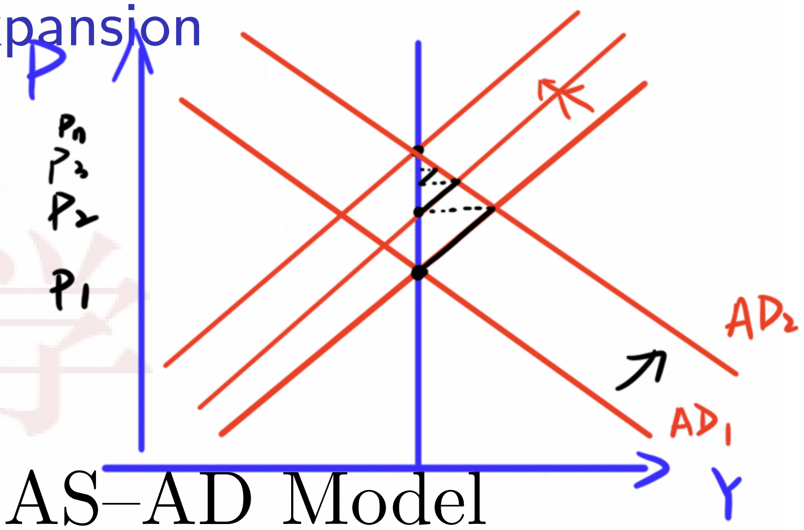
In the long run: $W_0 \searrow W_1 \Rightarrow N_1 \nearrow \overline{N}, \ Y_1 \nearrow \overline{Y}.$

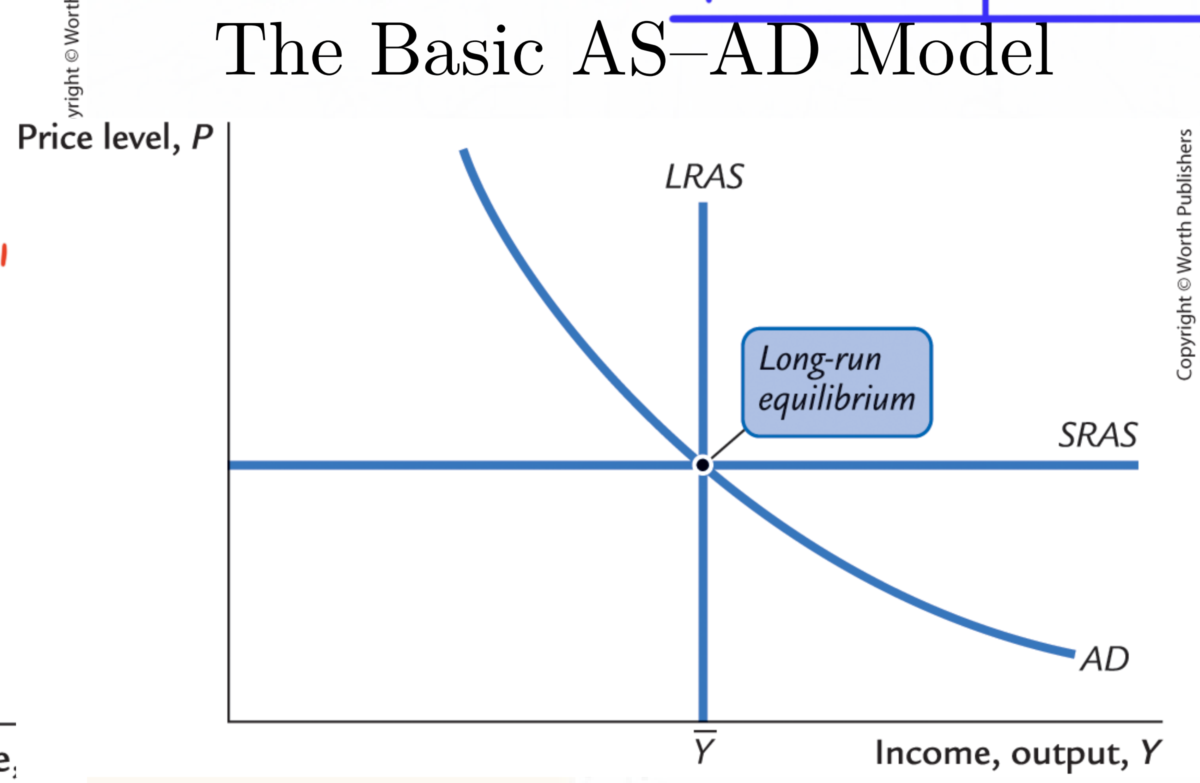
$$\overline{Y} = F(\overline{K}, \overline{N}), \quad Y_1 = F(\overline{K}, N_1).$$

The Standard AS–AD Model

The Short-Run and Long-Run Effects of Monetary or Fiscal Expansion

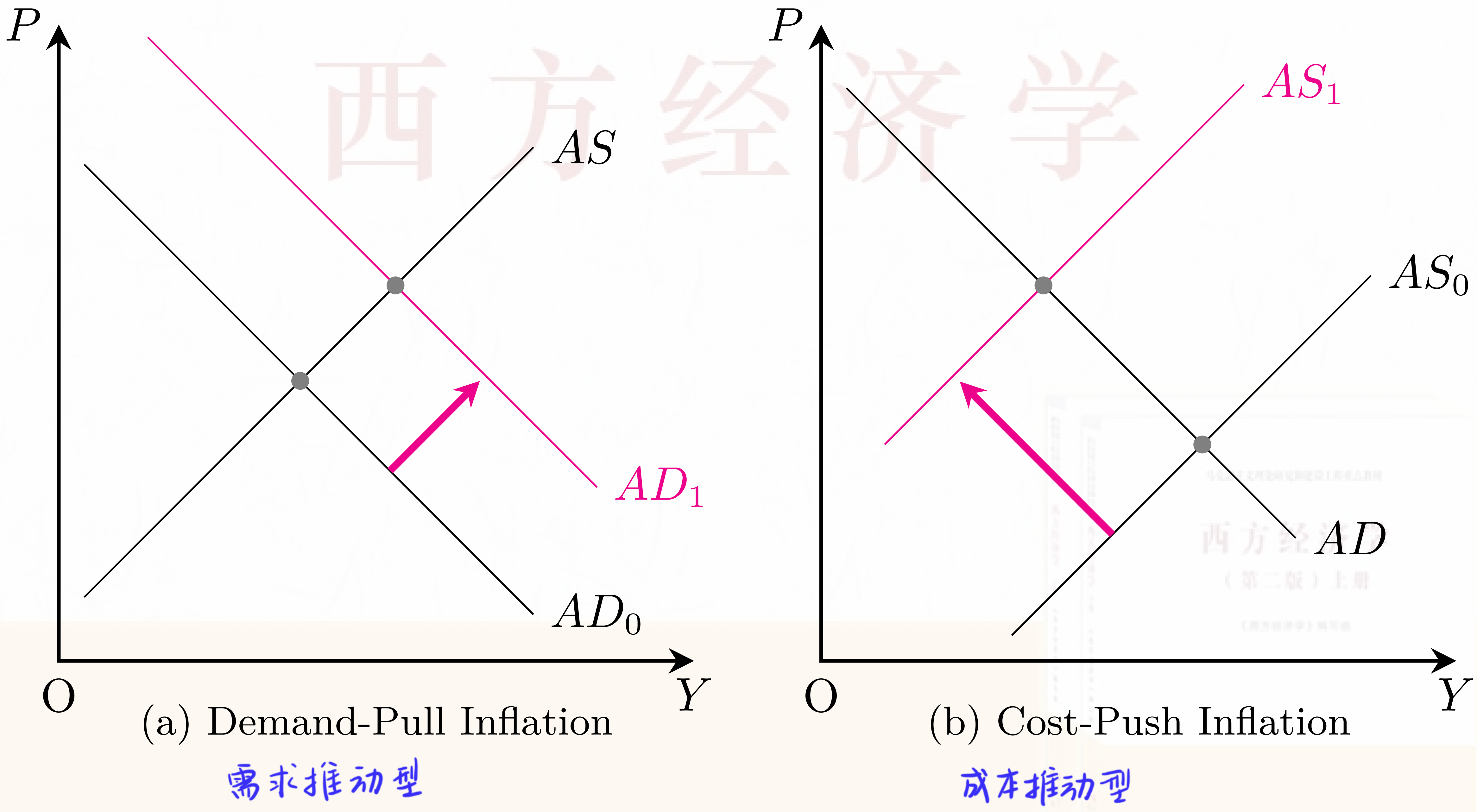






Outline

Two Causes of Inflation



Okun's Law in the United States

$x(t) = x(0)\, e^{gt}$

$x(t) = x(t-1)\, e^{g \cdot 1}$

$\Rightarrow \ln x(t) - \ln x(t-1) = g$

偏离百分比

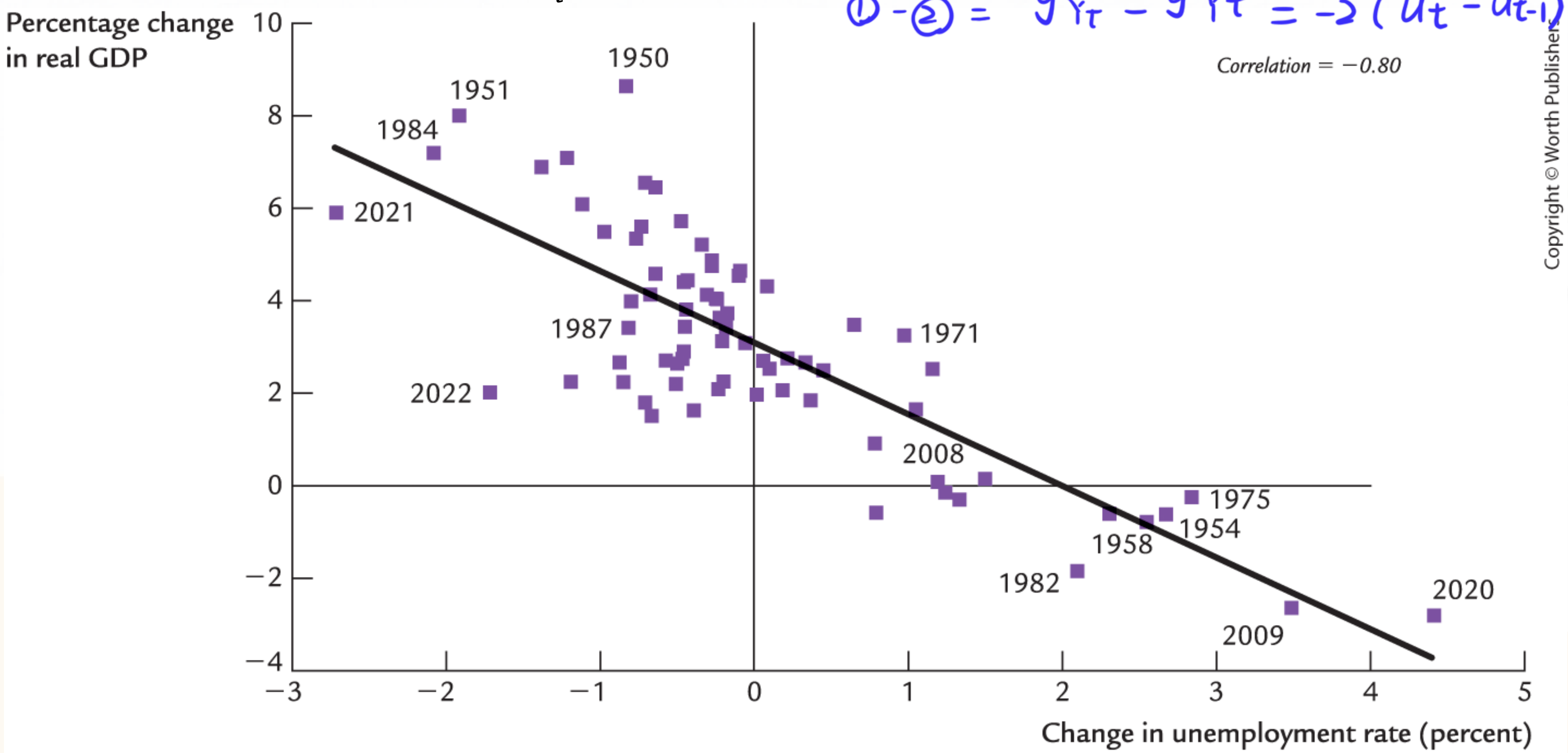
Okun's Law: ① $\ln(Y_t) - \ln(\overline{Y}_t) = -2(u_t - u_n)$.

The growth rate form: $g_{Y_t} - g_{\overline{Y}_t} = -2(u_t - u_{t-1})$.

② $\ln(Y_{t-1}) - \ln(\overline{Y}_{t-1}) = -2(u_{t-1} - u_n)$

① - ② = $g_{Y_t} - g_{\overline{Y}_t} = -2(u_t - u_{t-1})$

In the following figure, $g_{\overline{Y}_t} \approx 3$.



The Phillips Curve

$\ln x_t = \ln \bar{x} + \frac{x_t - \bar{x}}{x_t} + o(1)$

$Y_t = \bar{Y} + \alpha(P_t - E_{t-1}P_t)$

$Y_t - \bar{Y} = \bar{Y}(\ln Y_t - \ln \bar{Y})$

$= \alpha[\bar{p}(\ln P_t - \ln p) - E_{t-1}(P_t(\ln P_t - \ln \bar{P}))]$

$= \alpha[\bar{P}\ln P_t - E_{t-1}\bar{P}\ln P_t]$

$= \alpha\bar{P}[\ln P_t - E_{t-1}\ln P_t]$

By means of $X_t - \overline{X} \approx \overline{X}(\ln X_t - \ln \overline{X})$,

$$Y_t = \overline{Y} + \alpha \cdot (P_t - \mathbb{E}_{t-1} P_t)$$

$$\Rightarrow \ln P_t = \mathbb{E}_{t-1} \ln P_t + \frac{1}{\widetilde{\alpha}}(\ln Y_t - \ln \overline{Y})$$

$$\Rightarrow \pi_t = \mathbb{E}_{t-1} \pi_t + \frac{1}{\widetilde{\alpha}}(\ln Y_t - \ln \overline{Y}).$$

By Okun's law, $\ln Y_t - \ln \overline{Y} = -\widetilde{\alpha}\beta(u_t - u_n)$. Substituting for $\ln Y_t - \ln \overline{Y}$ gives

$$\pi_t = \pi_t^e - \beta(u_t - u_n),$$

which is called the **expectations-augmented Phillips curve**. If the supply shock v_t is considered, the Phillips curve becomes

$$\pi_t = \pi_t^e - \beta(u_t - u_n) + v_t,$$

短期粘性价格总供给曲线 AS原式

$$Y_t = \bar{Y} + \alpha(P_t - E_{t-1}P_t)$$

$\alpha > 0$ 未预期到物价上涨会拉高实际产出

$P_t > E_{t-1}P_t$（物价超预期上涨）$\Rightarrow Y_t > Y$ 产出高于潜在产出

$P_t < E_{t-1}P_t$（物价低于预期）$\Rightarrow Y_t < Y$ 企业收缩生产

$$P_t = E_{t-1}P_t + \frac{1}{\alpha}(Y_t - \bar{Y}) \quad ①$$

$$\ln(A + \Delta A) \approx \ln A + \frac{\Delta A}{A}$$

对①取对数

$$\ln P_t = \ln\left[E_{t-1}P_t + \frac{1}{\alpha}(Y_t - \bar{Y})\right] \approx \ln E_{t-1}P_t + \frac{Y_t - \bar{Y}}{\alpha \cdot E_{t-1}P_t}$$

$$\Rightarrow \ln P_t - \ln E_{t-1}P_t = \frac{Y_t - \bar{Y}}{\alpha \cdot E_{t-1}P_t}$$

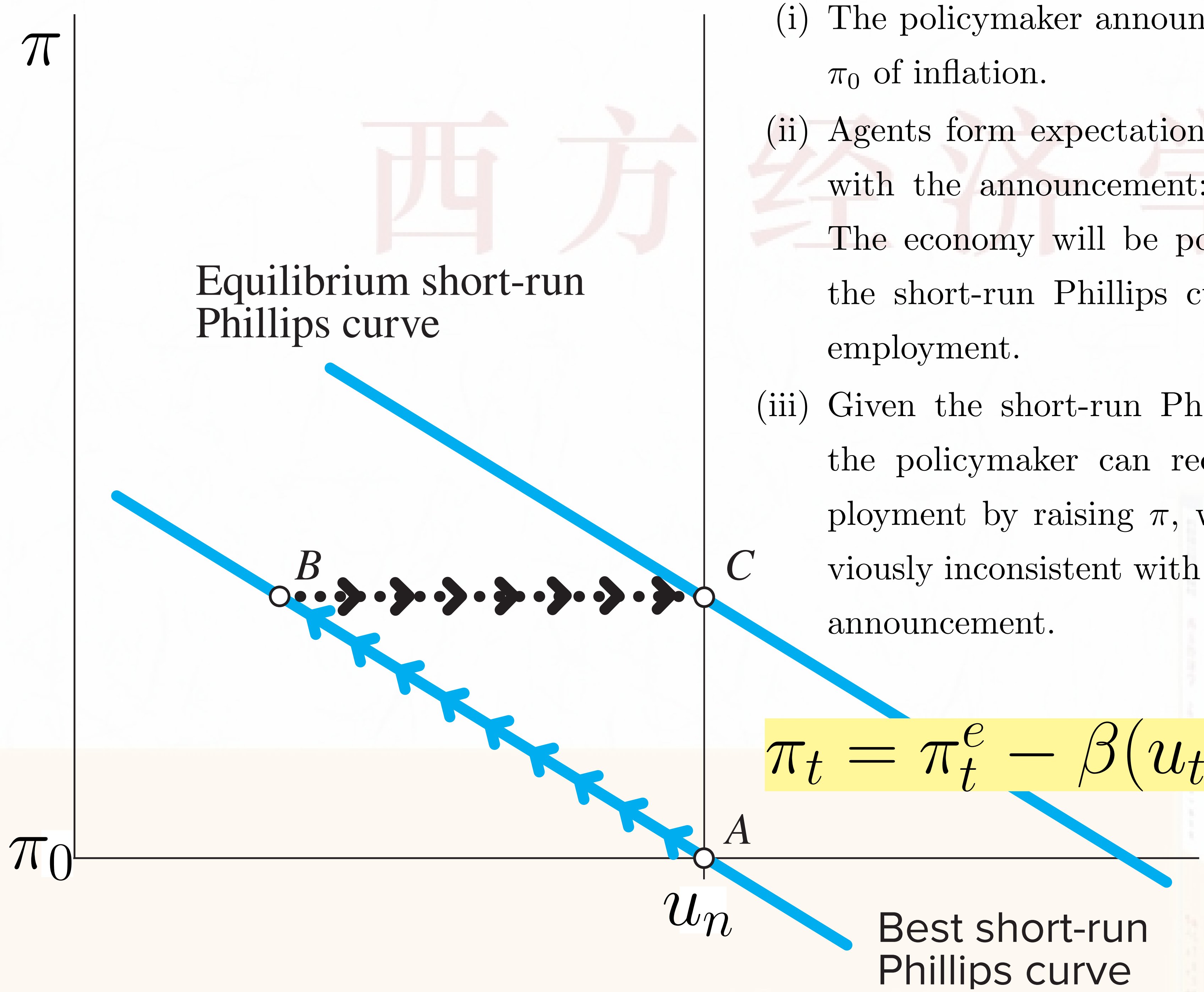
$$\ln P_t - \ln E_{t-1}P_t = \frac{Y_t - \bar{Y}}{\alpha \cdot E_{t-1}P_t}$$

$$\ln P_t - \ln E_{t-1}P_t = \frac{1}{\alpha}(\ln Y_t - \ln \bar{Y})$$

$$\pi_t - \pi_t^e = \ln P_t - E_{t-1}\ln P_t$$

The Phillips Curves

Dynamic/Time Inconsistency and the Long-Run Phillips Curve



(i) The policymaker announces a target π_0 of inflation.

(ii) Agents form expectations consistent with the announcement: $\pi^e = \pi_0$. The economy will be positioned on the short-run Phillips curve at full employment.

(iii) Given the short-run Phillips curve, the policymaker can reduce unemployment by raising π, which is obviously inconsistent with the original announcement.

$$\pi_t = \pi_t^e - \beta(u_t - u_n)$$

See Dornbusch (2018)

Rational Expectations

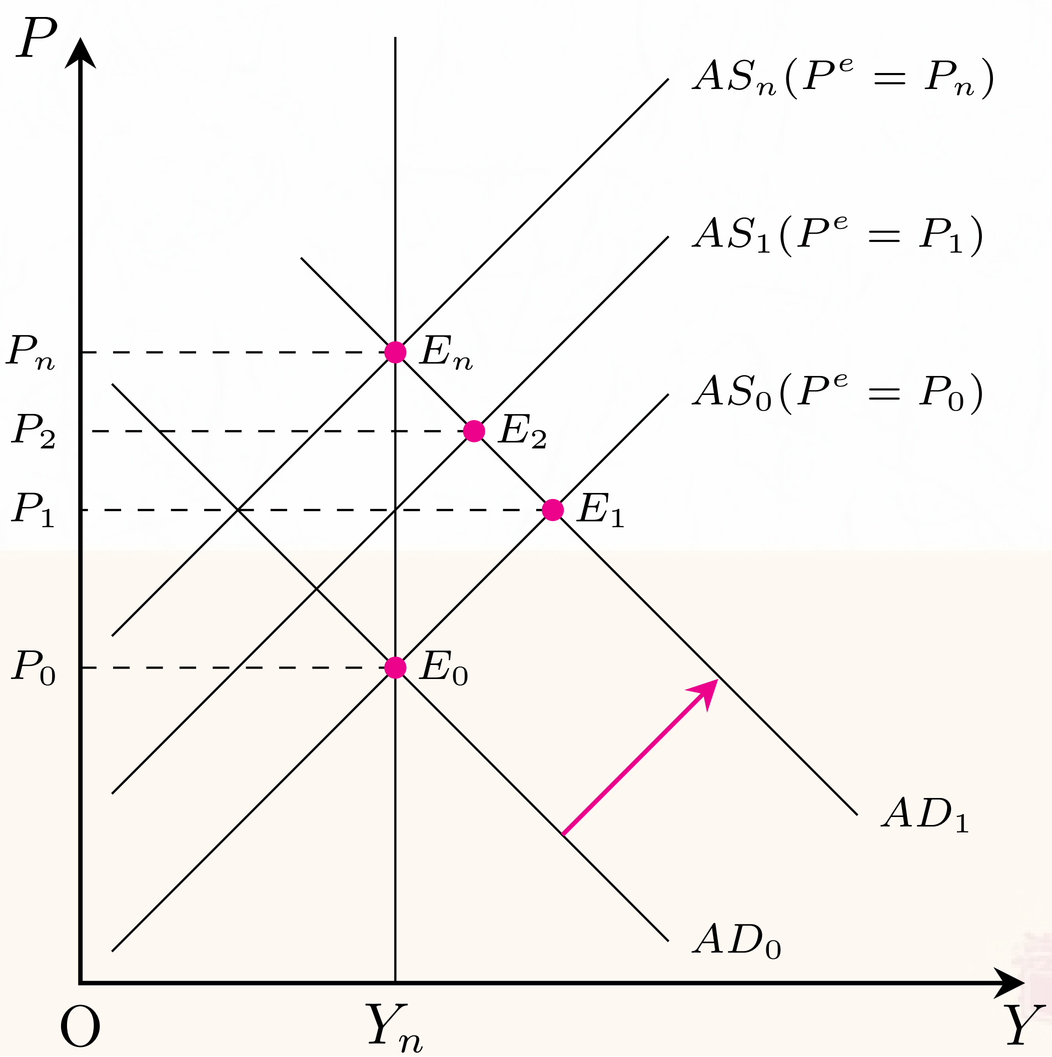
Adaptive expectations are given by $P_t^e = P_{t-1}^e + \lambda(P_{t-1} - P_{t-1}^e)$.

Rational expectations assume that

$\lambda=1$ 静态预期

(1) People inside a model know the model;

(2) People optimally use all the available information to forecast the future such that "outcomes do not differ systematically (i.e., regularly or predictably) from what people expected them to be."



Proposition 1 (Policy Ineffectiveness)

Under the assumption of rational expectations, policies expected by people have no effect on output.

Proposition 2 (Lucas Critique)

Under the assumption of rational expectations, forecasts based only on historical information are invalid.

Disinflation with Pain

Traditional Approach to Disinflation

The Phillips curve is

$$\pi_t = \pi_t^e - \beta(u_t - u_n).$$

Suppose that an economy is initially in full employment, but with high inflation π^{High}. Suppose the government plans to lower the inflation from π^{High} to π^{Target} during T years. Given π_t^e, π_t falls at the cost of high u_t. Pain caused by disinflation can be measured by the sacrifice ratio. [2] In terms of output lost, the sacrifice ratio is defined as the number of percentage points of one year's real potential GDP that must be forgone to reduce inflation by 1 percentage point.

$$\text{Sacrifice Ratio } (SR_Y) = \frac{\text{Output Lost}}{\text{Decrease in inflation}} = \frac{\sum_{t=1}^{T}(\ln \overline{Y} - \ln Y_t)}{\pi^{High} - \pi^{Target}}.$$

In terms of unemployment tolerated, it is defined as the number of percentage points of one year's cyclical unemployment that must be tolerated to reduce inflation by 1 percentage point.

$$\text{Sacrifice Ratio } (SR_u) = \frac{\text{Cyclical Unemployment Tolerated}}{\text{Decrease in inflation}} = \frac{\sum_{t=1}^{T}(u_t - u_n)}{\pi^{High} - \pi^{Target}}.$$

[2] Another measure is the misery index, defined as $u + \pi$ (Dornbusch, 2018, ch6).

Disinflation without Pain

Rational Expectations Approach to Disinflation

The central bank makes a credible announcement that the money supply will decrease immediately and the inflation target is $\pi_t = \pi^{Target}$. The public will form $\pi_t^e = \pi^{Target}$. According to the Phillips curve,

$$\pi_t = \pi_t^e - \beta(u_t - u_n).$$

As a result, $\pi_t = \pi^{Target}$ while $u_t = u_n$. The essential ingredient of successful disinflation is the credibility of monetary policy.

Case 1 Zimbabwe's inflation hit about 100% a day in 2008. The hyperinflation stopped by April 2009 because it is legal for everyone to use U.S. dollars. (See ch22 of Dornbusch [2018, 13th ed.]) 预期下降

Case 2 Forward guidance (See wiki, FED)

Taylor Rule

In order to establish the credibility of monetary policy, the central bank is advised to follow a set of prespecified and publicly announced rules. Such an example is the *Taylor rule*, [3] given by

$$i_t = \pi_t + r_n + \theta_\pi(\pi_t - \pi^{Target}) + \theta_Y \frac{Y_t - \overline{Y}}{\overline{Y}},$$

where $\theta_\pi > 0$ and $\theta_Y > 0$ are coefficients; i_t is the nominal interest rate controlled by the central bank; π_t is the rate of inflation; π^{Target} is the target rate of inflation; r_n is the natural rate of real interest; Y_t is the level of output; and $\overline{Y}$ is the natural level of output.

Proposition 3 (The Taylor Principle)

The central bank should respond to an increase in inflation with an even greater increase in the nominal interest rate. That is, $\frac{\partial i_t}{\partial \pi_t} > 1$.

Go to FRED to see the difference between effective federal funds rate and the rate implied by the Taylor rule.

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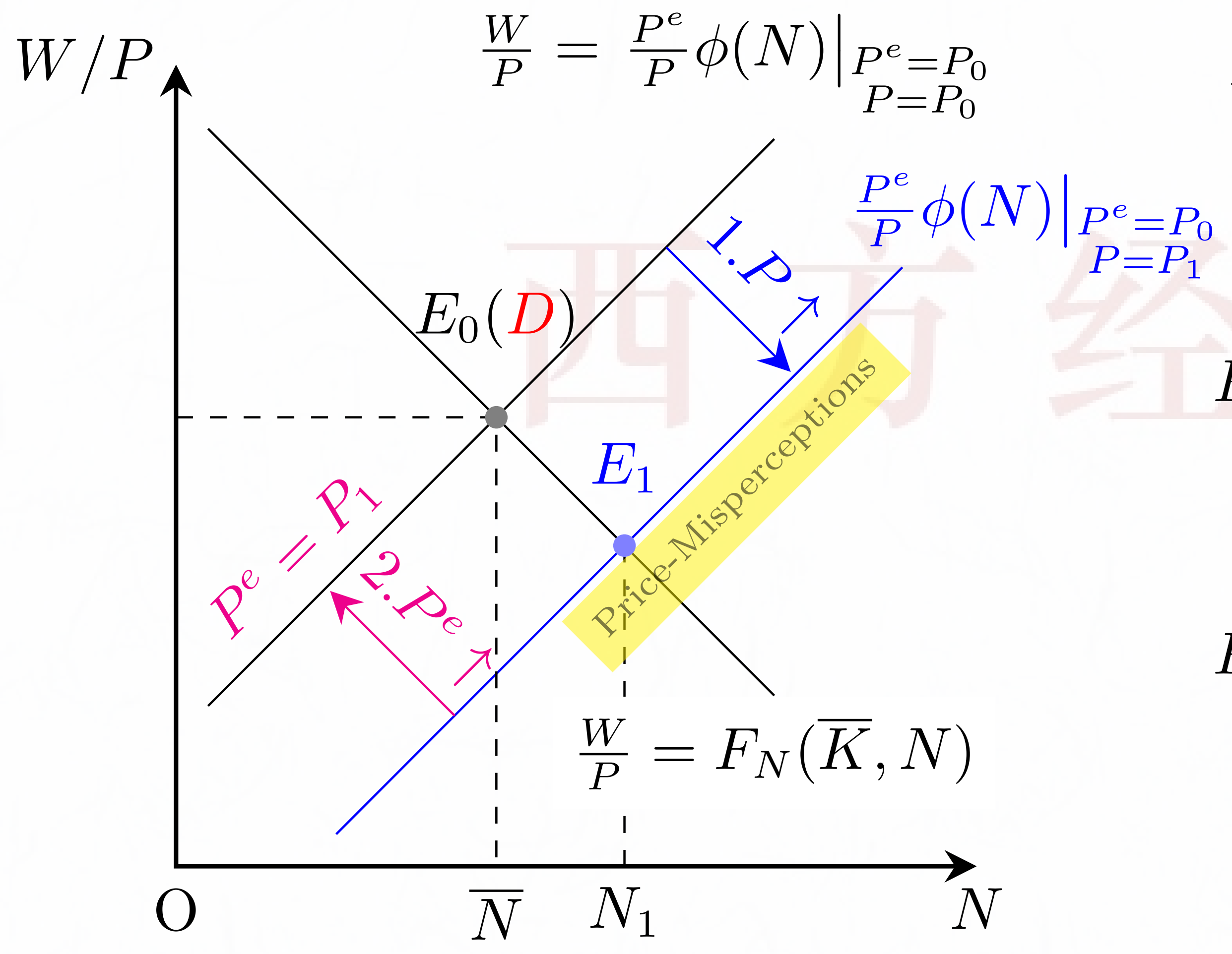
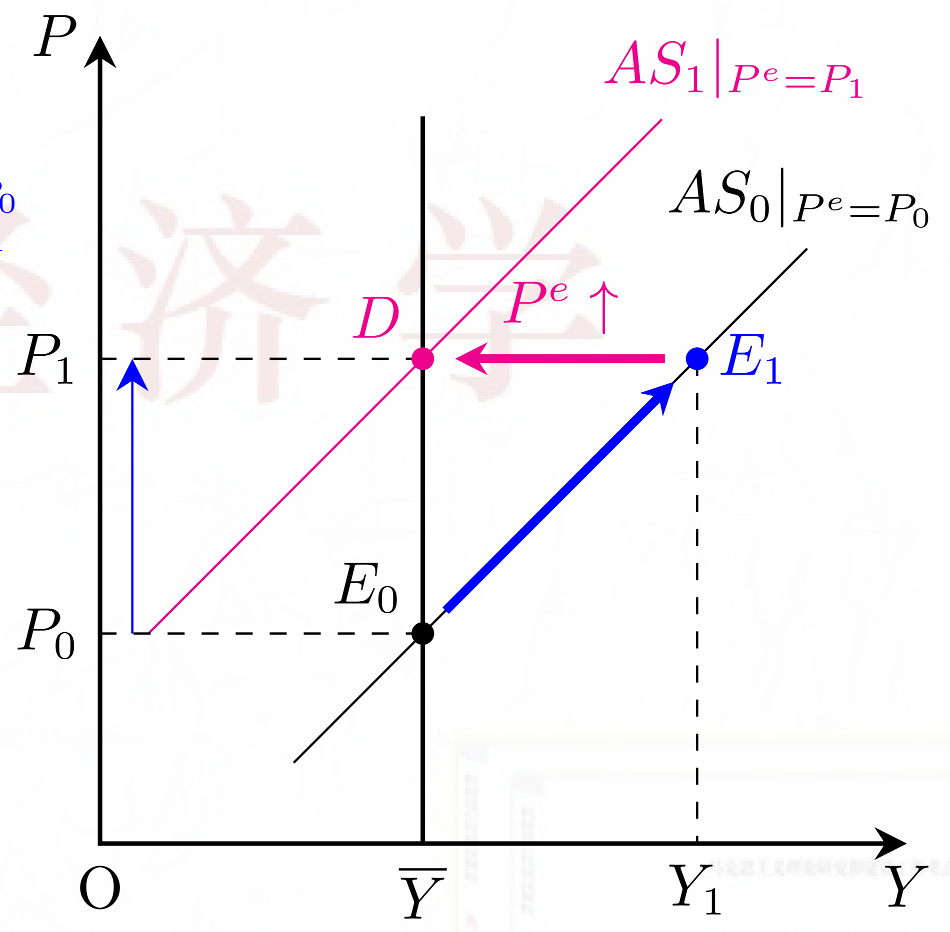
[3] John Taylor (1993, p.202) suggests a simple formula for monetary policy rule:

$$\text{The nominal Fed funds rate } i_t = \pi_t + 0.02 + 0.5 \times (\pi_t - 0.02) + 0.5 \times \left(Y_t - \overline{Y}\right)/\overline{Y}.$$

Outline

Appendix: The Workers' Price-Misperceptions Model (Friedman, 1968)





In the short run: $P_0 \nearrow P_1 \Rightarrow \overline{N} \nearrow N_1,\ \overline{Y} \nearrow Y_1.$

In the long run: $P^e \nearrow \Rightarrow N_1 \searrow \overline{N},\ Y_1 \searrow \overline{Y}.$

$$\overline{Y} = F(\overline{K}, \overline{N}), \quad Y_1 = F(\overline{K}, N_1).$$

Generally, the output is affected only by unperceived inflation.

Appendix: The Sticky-Price Model (Rotemberg, 1982)

When there are changes in demand due to changes in income, a fraction $s \in [0, 1]$ of firms charge sticky prices (P_{Sticky}) set in advance while a fraction $1 - s$ of firms charge flexible prices ($P_{Flexible}$). The general price level is $P = sP_{Sticky} + (1 - s)P_{Flexible}$. The price decision rules are

$$P_{Flexible} = P + a(Y - \overline{Y}), \quad a > 0;$$
$$P_{Sticky} = P^e + a(Y^e - \overline{Y}).$$

We assume that firms charging P_{Sticky} expect that the aggregate income, Y, will be equal to the natural level of output, $\overline{Y}$. Then the AS function can be written as

$$Y = \overline{Y} + \alpha(P - P^e)$$

where $\alpha = \frac{s}{(1-s)a} \geq 0$. AS is vertical if $s = 0$; horizontal if $s = 1$.

Appendix: The Imperfect-Information Model (Lucas, 1973)

There are n identical suppliers in an economy. Each supplier i can only observe one's own **flexible** price P_i. Due to imperfect information, each supplier cannot observe the general price level which is defined as $P = (\sum_{i=1}^{n} P_i)/n$ for simplicity. P^e is the expected general price level formed by supplier i based on all historical information. When supplier i observes his own price P_i, the expected general price level is revised to be $P^e|_{P_i}$.

$$\begin{aligned} \text{The update rule:} \quad & P^e|_{P_i} = P^e + b(P_i - P^e), \quad b \in [0,1]; \\ \text{The production decision rule:} \quad & Y_i = \overline{Y}_i + a(P_i - P^e|_{P_i}), \quad a > 0, \end{aligned}$$

where $\overline{Y}_i$ is the output corresponding to $b = 1$ under perfect information. Let $Y = \sum_{i=1}^{n} Y_i$ and $\overline{Y} = \sum_{i=1}^{n} \overline{Y}_i$. We obtain

$$Y = \overline{Y} + \alpha(P - P^e)$$

where $\alpha = a(1-b)n \geq 0$. This version of AS curve is called the *Lucas supply curve*.

Outline

疑难重点

(1) 掌握 AS 理论
(2) 掌握 AS-AD 模型，并分析政策效应。
(3) 什么是需求拉动型通胀？什么是成本推动型通胀？
(4) 什么是 Okun's law?
(5) 什么是 expectations-augmented Phillips curve?
(6) 理解适应性预期和理性预期的概念。
(7) 什么是 Lucas critique？什么是 policy ineffectiveness?
(8) 消除通胀有痛苦吗？有没有无痛消胀？什么是牺牲率？什么是前瞻性指引？
(9) 什么是 Taylor rule?

马工程教材疑难重点

1（E2, p.131）

根据马工程教材观点，应当如何评析西方经济学的 AS–AD 模型？

2（E2, p.177）

根据马工程教材观点，应当如何评析西方经济学的菲利普斯曲线理论？